Manuscript_1

Code and analysis pipeline for: *Passive Physiological Responses Fail to Predict Context-Dependent Action Selection in Bats*

> **Data availability:** Raw electrophysiology recordings and behavioral tracking files are deposited at [Zenodo]. See data below for details.

Overview

This repository contains all code used to analyze bat electrophysiology and behavioral data, including:

- An in vivo electrophysiology pipeline (spike sorting → response analysis → selectivity classification)
- Heart rate analysis during vocalization playback
- Video-based behavioral tracking analysis (approaches, time near speaker, latency)
- Statistical tests and publication-ready figure generation

Subjects were adult bats (R aegyptiacus) tested across four vocalization conditions: Aggression, Distress, Echolocation, and Distortion for passive listening recordings. For testing active listening cross contexts, Distress and White Noise were utilized in playback experiments. Animals were split by sex (male/female) and tested on Day 1 and Day 2 sessions.

Repository Structure

```
Manuscript_1/
├── Ephys.Pipeline.Final.ipynb     # Main ephys pipeline (Scripts 1–5 + supplementary cells)
├── BF_Heatmap_Final.ipynb         # Best frequency heatmap (binary BF visualization)
├── Nonparametric.R        # Heart rate statistical analysis (R)
├── Hr.Example.py              # Heart rate timeseries plots (per stimulus, by sex)
├── RealRadar.Nonpara.py         # Heart rate radar chart + nonparametric stats
├── BORISGraphs.py             # Behavioral bout analysis from BORIS annotations
├── NEW_ST.py                  # Speaker approaches + time near speaker (4 conditions)
├── ST_overview.py             # Two-part approach analysis (proportion + metrics)
├── SpeakerTestHeatMap.py         # KDE position heatmaps across conditions
├── approaches_day1d2.py         # Day 1 vs Day 2 approach comparison
└── xpos_ST.py                # Directional movement analysis (toward/away speaker)
```

Data

Data types

Type	Format	Description
Electrophysiology recordings	.npy (continuous)	Raw spike-sorted continuous traces, one file per animal
Ephys per-call output	.npy	Produced by Script 1; contains spike rates, waveforms, timestamps per call
Ephys category summaries	.npy	Produced by Script 2; averaged per animal × vocalization category

Type	Format	Description
Heart rate timeseries	.csv	One file per bat per session; columns: Time, Heart_Rate
Behavioral tracking	.csv	DeepLabCut/tracking output; columns include frame-by-frame x, y position
BORIS behavioral annotations	.csv	Ethogram export with Time, Behavior, Behavior type (START/STOP)

Data access

Raw data are deposited at **[Zenodo]**:

Expected local directory structure (update paths in each script before running):

```
data/
├── ephys/
│   ├── continuous/         # Raw .npy continuous recording files (one per animal)
│   └── processed/          # Auto-generated by Ephys.Pipeline.Final.ipynb
│       └── Bat_ID/
│           └── category/   # Script 1 + Script 2 output
├── heart_rate/
│   ├── Females/            # HR .csv files for female bats
│   └── Males/              # HR .csv files for male bats
├── tracking/               # Video tracking .csv
│   ├── Male.Videos/
│   └── Female.Videos/
└── boris/                  # BORIS annotation .csv files
```

Usage

1. Electrophysiology pipeline — Ephys.Pipeline.Final.ipynb

Run cells in order. Each cell is a self-contained script:

Script	What it does	Input	Output
Script 1	Per-call processor	Continuous .npy files	Per-call .npy files
Script 2	Category averager	Per-call .npy files	Category-average .npy files
Script 3	Heatmap + Random Forest selectivity	Category .npy files	Publication figures
Script 4	Latency + spike jitter analysis	Per-call .npy files	Latency/jitter figures
Script 5	Raster + PSTH + waveform plots	Continuous + per-call .npy	Per-animal figure PDFs
Supp. cells	Selectivity heatmap, sparseness scatter, GLM decoder, violin plots	Category .npy files	Additional figures

Before running: update INPUT_FILE_PATHS and output directory paths at the top of each script cell.

2. Best frequency heatmap — BF_Heatmap_Final.ipynb

Standalone notebook. Update INPUT_FILE_PATHS to point to your ephys data directory. Produces a binary heatmap showing best-frequency tuning across the recorded population.

3. Heart rate analysis

R script (Nonparametric.R): primary statistical analysis. Computes robust baseline HR, calculates change scores per stimulus window, runs Kruskal-Wallis + Dunn post-hoc tests, and produces ggplot2 figures. Update female_folder_path and male_folder_path at the top.

Python scripts: - Hr.Example.py — plots mean ± SEM heart rate timeseries per stimulus type, averaged by sex - RealRadar.Nonpara.py — radar chart of HR change per vocalization category + Mann-Whitney U comparisons

4. Behavioral analysis

All scripts share a similar CONFIG block at the top. Update BASE_DIR to your tracking CSV directory before running.

Script	Analysis	Output
BORISGraphs.py	Behavior bout durations from BORIS annotations	Bar/violin plots per behavior
NEW_ST.py	Approaches + time near speaker across 4 conditions	Bar plots with significance stars
ST_overview.py	Proportion of approachers + latency/duration among approachers	Two-panel figure
SpeakerTestHeatMap.py	KDE position heatmaps per condition	2×2 heatmap grid
approaches_day1d2.py	Day 1 vs Day 2 approach counts per condition	Grouped bar plot
xpos_ST.py	Signed x-displacement during stimulus (toward/away from speaker)	Boxplot per condition

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Requirements

Python

Tested with Python 3.9+. Install dependencies with:

```
pip install numpy pandas matplotlib seaborn scipy scikit-posthocs statsmodels
```

Full dependency list:

```
numpy
pandas
matplotlib
seaborn
scipy
scikit-posthocs
statsmodels
pathlib      # stdlib
glob         # stdlib
itertools    # stdlib
```

> **Note:** Ephys.Pipeline.Final.ipynb and BF_Heatmap_Final.ipynb require Jupyter.

R

Tested with R v2024.04.0, Install packages with:

```r
install.packages(c("ggplot2", "dplyr", "viridis", "ggpubr", "FSA", "lme4", "lmerTest"))
```

Animals

Individual bats are identified by short alphanumeric IDs (e.g., AF4, B07, AEO). Sex assignments are hardcoded in RealRadar.Nonpara.py and Hr.Example.py — see MALE_BATS sets therein. Stimulus timing windows differ for a subset of animals (AF4, AF6, B32, AD6) due to session timing differences; alternative windows are applied automatically in Nonparametric.R.

Citation

If you use this code, please cite:

License

Contact

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